

Kisan ISP - External Payment Public API Guide

Production Documentation & Integration Manual | Version 1.0.0

1. Overview & Architecture

The External Payment Gateway (externalpayment) provides an enterprise REST API that allows third-party payment gateways, bank channels, kiosk aggregators, and custom portals to push instant subscriber recharges, query customer balances, and synchronize billing transactions automatically with Kisan ISP.

It mirrors 100% of the proven architecture from existing payment integrations (such as eSewa) while introducing direct push recharge capabilities where an external service can recharge an account directly using a customer username and duration without requiring a prior inquiry request.

2. Endpoints Summary

POST /api/externalpayment/access-token	Obtains OAuth2 Bearer access token using username & password
POST /api/externalpayment/login	Direct JSON login alternative returning Bearer token
GET /api/externalpayment/inquiry/:id	Search user by PPPoE username, customer ID, phone, or email
GET /api/externalpayment/user/:username	Lookup customer directly by PPPoE username (e.g. karnkalyan)
POST /api/externalpayment/payment	Direct Push Recharge: renews subscription, RADIUS & invoice
POST /api/externalpayment/recharge	Alias to /payment for direct recharge push
POST /api/externalpayment/status	Transaction reconciliation & status query by transaction_code
GET /api/externalpayment/payment-modes	List available payment methods (EXTERNAL, CASH, ONLINE, etc.)
GET /externalpayment	Frontend management dashboard & instant recharge tester

3. Authentication Methods

The API supports multiple authentication strategies to ensure seamless compatibility with external servers, cron jobs, and webhooks:

Bearer Token	Header: Authorization: Bearer <access_token> (expires in 24 hours)
HTTP Basic Auth	Header: Authorization: Basic <base64(username:password)>
Payload Credentials	Body: "auth_username" & "auth_password" (or "username" & "password")
API Key	Header: x-api-key: <api_key>

External Payment API - Customer Inquiry & Recharge

Step-by-Step API Specification & Direct Recharge Workflows

4. Customer Inquiry (Inquiry API)

Query customer status, current subscribed package, plan expiration date, and all available online packages for renewal. The inquiry parameter accepts: PPPoE username (e.g. karnkalyan), Customer Unique ID (e.g. CUST-1001), mobile phone number, or email.

HTTP Request:

```
GET https://cms.kisan.net.np/api/externalpayment/inquiry/karnkalyan
Authorization: Basic ZXh0ZXJyYXxfXNwXzE6RXh0ZXJyYXxASVNQIyExMjAyNQ==
```

Success Response (JSON):

```
{
  "response_code": 0,
  "response_message": "success",
  "customer": {
    "id": 142,
    "customer_unique_id": "CUST-1001",
    "customer_name": "Kalyan Karn",
    "username": "karnkalyan",
    "phone": "9800000000",
    "expiry_date": "2026-10-01",
    "status": "active",
    "is_rechargeable": true
  },
  "current_package": {
    "id": 12,
    "name": "KISAN-75-1-TV",
    "duration": "1 month",
    "amount": 1200
  },
  "packages": [
    { "id": 12, "name": "KISAN-75-1-TV", "duration": "1 month", "amount": 1200 },
    { "id": 13, "name": "KISAN-75-3-TV", "duration": "3 months", "amount": 3400 }
  ]
}
```

5. Direct Push Recharge API

The primary endpoint for pushing payment confirmation and instant recharge. When called, the system automatically resolves the customer, calculates the new expiration date, provisions RADIUS, syncs the sales invoice to accounting, and marks the account active.

HTTP Request (cURL):

```
curl -X POST "https://cms.kisan.net.np/api/externalpayment/payment" \
-H "Content-Type: application/json" \
-u "external_isp_1:External@ISP#1!2025" \
-d '{
  "username": "karnkalyan",
  "payment_mode": "EXTERNAL",
  "duration": "1 month",
  "amount": 1200,
  "transaction_code": "EXT-TXN-9847291"
}'
```

Success Response (JSON):

```
{
  "response_code": 0,
  "response_message": "Payment and recharge successful",
  "data": {
    "transaction_code": "EXT-TXN-9847291",
    "reference_code": "EXT-20260916-B97FC73659FE-24",
    "order_id": 24,
    "customer_id": "CUST-1001",
    "customer_name": "Kalyan Karn",
    "username": "karnkalyan",
    "payment_mode": "EXTERNAL",
    "amount": 1200,
    "package_name": "KISAN-75-1-TV",
    "package_duration": "1 month",
    "new_expiry_date": "2026-11-01T18:15:00.000Z",
    "radius_provisioned": true
  }
}
```

External Payment API - Network & Accounting Sync

RADIUS Provisioning, Accounting Integration & Reconciliation

6. RADIUS Provisioning & Network Refresh

Upon every successful recharge, the system provisions the RADIUS AAA database in real-time:

Expiration Attribute	Updated in radreply table to the exact new calculated package end date.
Session Disconnection	Sends a disconnect request (CoA / PoD) to NAS/BRAS so the user immediately reconnects with the renewed speed profile and updated quota.
Service Status	Activates connection user status and sets customer service connection to active.

7. Accounting Invoice Synchronization

Every external payment automatically generates an official sales invoice in the configured cloud accounting system (Tshul Accounting or Nepurix Cloud Accounting):

Customer Record	Matches or creates customer ledger in accounting using customer unique ID.
Line Items	Breaks down Base Package, VAT (13%), Telecommunication Service Charge (TSC), and add-on charges.
Payment Method	Allocated to the specified payment_mode (e.g. EXTERNAL, CASH, ONLINE) in accounting books.

8. Status Check & Reconciliation API

Check status of any transaction to reconcile pending payments or confirm transaction execution.

HTTP Request:

```
POST https://cms.kisan.net.np/api/externalpayment/status
Content-Type: application/json
Authorization: Basic ZXh0ZXJuYXZfaXNwXzE6RXh0ZXJuYXZASVNQIyExMjAyNQ==

{
  "transaction_code": "EXT-TXN-9847291"
}
```

Response:

```
{
  "response_code": 0,
  "status": "COMPLETED",
  "response_message": "success",
  "data": {
    "transaction_code": "EXT-TXN-9847291",
    "reference_code": "EXT-20260916-B97FC73659FE-24",
    "amount": 1200,
    "payment_mode": "EXTERNAL",
    "paid_at": "2026-09-16T10:07:45.000Z"
  }
}
```

External Payment API - Access, Setup & Credentials

Credential Management, System Settings & Dashboard Access

9. ISP Credential Generation & Management

Each ISP tenant can independently generate and customize their external payment API username and password. This ensures complete isolation between ISPs.

Master Settings Tab	System Settings -> Master Settings -> External Payment tab.
Dedicated Portal	Services -> 3rd Party Services -> External Payment API Requests.
Credential Generator	Click "Generate Random Password" or enter a secure password and save.
Default Payment Mode	Configurable per ISP (default: EXTERNAL).
API Authentication	Supports HTTP Basic Auth (username:password), Bearer Token, or JSON body.

10. Frontend URLs & Live Recharge Tester

Dashboard URL	https://cms.kisan.net.np/externalpayment
Alternative URL	https://cms.kisan.net.np/services/externalpayment
Sidebar Navigation	Services -> 3rd Party Services -> External Payment API Requests
Live Tester Tab	Allows operators to enter a username & duration to test live push recharges directly.